



## **Prompt-Data Analysis**

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A capstone project report submitted to the faculty of  
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## WORD DOC formatting:

## Roles

## Resumes

1. Each section of the report must begin on a new page.
2. Sections should be numbered according to the Report Structure given in the CapStone system.

Font in level-1 headings should be 14 point, bolded.

3. Font in level-2 headings should be 14 point.
4. Font in level-3 headings should be 12 point, bolded.
5. Font in level-4 headings should be italic 12 point.
6. Lines of text within a paragraph should be single spaced. Use 1.5 line spacing between paragraphs
7. Headings and paragraphs should NOT be indented.
8. Figures in the text should be numbered consecutively. Underneath each figure should be a label with its index number and caption (12 point, bolded). The figure numbers should be used to refer to figures in the body of text.
9. Tables are given in the appendices. The table numbers should be used to refer to tables in the body of the text.

**Computer Information Systems Capstone Report Structure**

1. Abstract(Ryan) [200-300 words]

The rapid adoption of generative artificial intelligence has outpaced many users ability to interact with these systems effectively, creating a growing gap in prompt literacy. At California State University San Marcos, students and faculty are provided free access to the latest OpenAI models through an institutional license. However, many users are unsure how to write prompts that fully take advantage of these tools. This project proposes the development of a campus-wide prompt sharing website where students and faculty can create, explore, and share effective AI prompts. By supporting a collaborative collection of prompts, the platform aims to improve AI literacy, encourage responsible and creative use of generative AI, and reduce barriers for new users. The main goal is to foster a space that welcomes prompt sharing for the purpose of learning and inspiring through AI leveraging. The platform will include prompt descriptions, intended use cases, and examples of expected outputs, allowing users to understand not only what a prompt does, but why it works. Prompts may be tagged by subject or task type to support discovery. Overall, the website is intended to foster peer learning, experimentation, and sharing best practices for using AI in an academic setting to learn more.

2. Report Revision History
  1. Changes in Version 2.0
  2. Changes in Version 3.0

### 3. Problem Statement

#### 1. Business Context (Devin)

Prompt Marketing is a new market in the AI, Tech sector. A multitude of emerging prompt market websites are geared to create a transactional prompt buying and selling atmosphere where the main incentive is money. Creators of prompts sell their prompts for a profit, while consumers spend money to gain access to curated prompts. Existing websites categories hundreds to thousands of prompts for users to search through and some implement a search or filter or an enhanced experience. A main purpose identified with the current websites is generating revenue from prompt transactions. As AI creeps into various sectors, the need for writing effective and quick serving prompts is becoming a necessity. The use of AI, although critiqued by some, will continue to grow exponentially undeniably creating a large market for AI products and services.

Our website as a competitor to existing sites will have a competitive advantage by providing a free platform for prompt exchange. The growth of the site will be credited to its overall purpose which can be defined as a space for prompt searches built through community engagement.

#### 2. Business Problem (Angelina)

Educational institutions rely on written content for communication and instruction, like event flyers, homework assignments, announcements, rubrics, and instructional materials. However, both students and professors sometimes come to a halt on creating quality, clear, and engaging content continuously. Students may lack experience with professional writing and formatting, while professors face time constraints and repetitive workload when creating similar materials for each term.

The current tools in existence provide basic templates that often feel disconnected from the academic context and do not show the needs, language, or creativity of actual students and educators. As a result, the users end up spending more time than necessary for rewriting, reformatting and searching for examples that match the specific characteristics that best represent the school.

The main business problem is the absence of a centralized, trusted platform where students and professors can create, share, discover, and reuse quality prompt templates that specifically represent the school's academic and campus needs. Without this platform, productivity is reduced, and knowledge/creativity maintains the disconnect.

Prompt Market aims to solve this problem by providing a marketplace style platform where students and professors can contribute and access peer created prompt templates for events, flyers, assignments, announcements, and other academic needs. This platform improves not only consistency but quality of content and efficiency while allowing us to work together as an educational community.

#### 3. Objectives

Objectives include designing and developing a free, user-friendly AI prompt platform that allows students to create, like, dislike, comment, browse and discover helpful and creative prompts. Our website will focus on promoting high-quality and effective

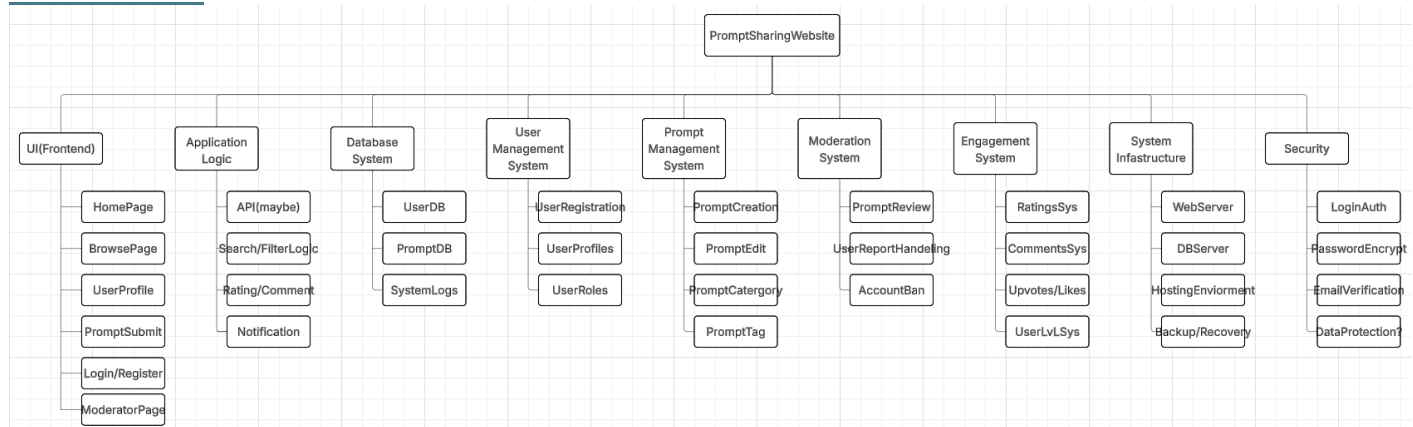
prompts aimed at improving user productivity and learning outcome when using AI tools. User engagement will be supported through a community-built site. Users will engage with prompts through comments and likes and will have ranks, levels for engagement as well. This form of incentive will produce quality prompts and will keep visitors on the site longer.

#### 4. Project Management

##### 1. Project planning

##### 1. Product Breakdown Structure (Ryan)

[https://lucid.app/lucidchart/0721729d-cc57-4e20-843e-85a10077389e/edit?viewport\\_loc=-1449%2C-963%2C3518%2C1778%2C0\\_0&invitationId=inv\\_08573e4e-647d-4589-b76a-9742b0433180](https://lucid.app/lucidchart/0721729d-cc57-4e20-843e-85a10077389e/edit?viewport_loc=-1449%2C-963%2C3518%2C1778%2C0_0&invitationId=inv_08573e4e-647d-4589-b76a-9742b0433180)



##### 2. Work Breakdown(Luis)

The project is divided into important functional components. This includes the user requirements, functional requirements, and non-functional requirements, as well as the methods of construction for the system. The project's main functional components include the frontend design component, the backend system design, the database structure, and finally the deployment. This includes user requirements, defined functional requirements, and defined non-functional requirements. User requirements describe what the users expect from the system. Functional requirements define what the system must do, such as authentication, search functionality, and CRUD operations. Non-functional requirements describe system qualities such as performance, usability, reliability, and security. The system design includes Canva conceptual diagrams, UML designs, ERDs, and UI wireframes. These designs serve as the blueprint for the system. The UML diagrams model system structure and relationships, the ERDs define the database schema and data relationships, and the UI wireframes outline how the interface will appear and function.

Frontend development includes the login and registration pages, the browse and search functions, the AI prompt description pages, search filters, and the moderation/admin page. The frontend focuses on the user interface and user experience, ensuring that users can easily navigate, search, and interact with prompts on the platform.

Backend development includes CRUD operations, the authentication system, API development, and server-side logic using Node.js and a backend framework. The backend handles data processing, user validation, database interactions, and enforcement of system rules. It ensures secure login, proper data storage, and communication between the frontend and database.

It is further divided into components where each member is assigned essential tasks to ensure accountability and completion of their part of the system requirements.

### 3. Gantt Chart(Marlene)

#### 2. Cost-Benefit Analysis

##### 1. Cost Analysis (17\$/hr)(Devin)

The development of the AI Prompt Marketplace app will have several upfront and operational financial investments. The initial developmental costs will include building the frontend, backend and database infrastructure of the app. This will come with an estimated cost range of 10k depending on the size and experience of the development team. There will also be a cost of a security implementation which would cost an add on of 1-2k. The total upfront cost to start up this project will be roughly \$12,000. Once the platform is now operational, there will be a monthly expense to keep up the project. Cloud hosting services and database storage are expected to cost \$1k per month, while there is ongoing maintenance and technical support that will add another 1500-2k per month. The platform will also have a cost of 10-20 dollars per year. Overall we expect a one time initial investment of \$12,000 and operational costs every month of \$3000.

### 2. Benefit/Impact Analysis(Angelina)

#### 3. Progress review

#### 5. Requirements (Luis)

##### 1. Original User Requirements

##### 1. Glossary of Relevant Domain Terminology

| Term               | Definition |
|--------------------|------------|
| AI                 |            |
| Prompt engineering |            |
| Prompt Market      |            |
| AI engines         |            |
|                    |            |
|                    |            |

### User Groups

Users are the primary participants in the system. They can browse and search for prompts, view prompt details, provide feedback through comments and ratings, and upload or manage their own prompts. Users act as both consumers and contributors within the marketplace.

Contributors are users who choose to upload prompts to the platform. They can create, edit, and update prompts, and may earn levels based on community engagement such as upvotes and downloads. This role is not separate from users but represents a higher level of participation.

Moderators oversee platform activity to ensure quality and appropriate use. They review submitted prompts, approve or reject content, handle reports of bad user or interaction, and manage user accounts.

## **2. Functional Requirements**

- Users must be able to browse and search for prompts within the marketplace
  - Enables users to discover relevant prompts efficiently
- Users must be able to view detailed prompt information
  - Includes descriptions, categories, and example outputs
- Users must be able to rate prompts and provide feedback through comments
  - Allows community-driven evaluation of prompt quality

### **Contributors**

- Contributors must be able to upload new prompts to the platform
  - Allows users to share AI prompts with the community
- Contributors must be able to edit or update their uploaded prompts
  - Ensures prompt accuracy and continuous improvement
- Contributors must be able to track engagement metrics
  - Metrics include upvotes, downloads, and ratings

### **Moderators**

- Moderators must be able to review submitted prompts before or after uploading
  - Ensures compliance
- Moderators must be able to approve, reject, or remove prompts
  - Applies when prompts violate platform guidelines
- Moderators must be able to manage user accounts
  - Includes issuing warnings, suspensions, or removals

## **4. Non-functional Requirements [Short Descriptions of Each type of Non-Functional User Requirements given in Appendix R]**

### **1. Product: Usability Requirements**

- =The user interface must be smooth and easy to use for all users
  - must be easily for first-time users
- The system must present prompts and search results in a clear, organized layout
  - Improves readability and usability
- The system must provide immediate and understandable feedback
  - For actions such as searches, ratings, and comments
- The system must provide a clear interface for content management
  - Applies to contributors and moderators

## 2. Product: Performance Requirements

- The system must return search results without noticeable delay.
- The system must allow ratings and comments to update in real time.
- The system must handle multiple users without too much lag

## 3. Product: Availability/Reliability/Security

The system must be available to users

- Supports constant access

The system must ensure reliable access to uploaded and saved prompts

- Prevents data loss

The system must protect user accounts and uploaded content

- Prevents unauthorized access

## 4. Other Constraints

- The system must be accessible through standard web browsers

### 2. Refined System Requirements

#### 1. Functional Requirements

##### 1. System Functional Requirements [Short Descriptions of System Functional Requirements given in Appendix R]

- System must validate login credentials
  - Validation of username and password is necessary for successful entry
- System must validate user emails
  - System must validate and verify legitimate emails during user registration
- System must store user/prompt data in a database
  - Data includes: user information (username, email, password, comments, posts, upvotes, downvotes, level), prompt information (title, description, category, comments, upvotes, downvotes, downloads, ratings), user reports(warnings)
- System must retrieve search results from the database
  - System will use category search to return requested prompt
- System must be able to suspend/terminate accounts
  - Users requesting account termination will have their accounts deactivated by the system

#### 2. Data/Information Requirements

- System must store user data, prompt data, and system data
- User data: UserID, fullname, email address, password, account status,
- Prompt data: PromptID, title, description, category, tags, ratings, upvotes, downvotes, downloads
- System data: user levels, saved prompts, login logs,

##### 2. Non-functional Requirements [Short Descriptions of Each type of Non-Functional System Requirements given in Appendix R]

**1. Product: Usability Requirements**

- System must provide a user-friendly interface
- System must maintain a consistent layout & navigation across all pages
- System must provide clear messages and feedback
- System must allow users to complete registration in <3 minutes

**2. Product: Performance Requirements**

- System must load all pages within 3 seconds
- System must process search requests <5 seconds
- System must support  $\geq 2,000$  concurrent users

**3. Product: Availability/Reliability/Security**

- System must maintain 99% uptime
- System must recover from system failure within 10 minutes
- System must encrypt sensitive user data
- System must enforce password strength requirements

**4. Other Constraints**

- System shall be implemented using HTML, CSS, JavaScript & backend framework
- System database will use MySQL
- System will support 24/7 operation

3. Requirements Trace Table [Short Descriptions of the Requirements Mapping table given in Appendix R]

**6. Exploratory Studies(Luis)**

1. Relevant Development Frameworks [frameworks, or technology stacks, upon which your system can be built; at least one page]
2. Relevant Solution Techniques [3rd-party open-source packages and libraries that may be used as part of the solution, and specific algorithms/techniques required by the project; at least one page]

**7. System Design**

1. High-level Design(Ryan)
2. Low-level Design [UML Class Diagrams; ERD for DB]
3. User Interface Design

**8. System Implementation (Luis)**

1. Programming Languages & Tools
2. Coding Conventions
3. Code Version Control
4. Implementation Alternatives & Decision Rationale

**9. System Testing Report****10. Challenges & Open Issues****1. Challenges Faced in Requirements Engineering(Marlene)**

- [Availability of industry mentor]
  - [Understanding the problem domain]
2. Challenges Faced in System Development
    - [What new techniques you need to learn]
    - [Do you have sufficient tool support]
    - [Any difficulties in agile practices]



3. Open Issues & Ideas for Solutions
11. System Manuals (Luis)
  1. Instructions for System Development
    1. **How to set up development environment**
    2. **Notes on system further extension**
  2. Instructions for System Deployment
    1. **Platform Requirements**
    2. **System Installation**
  3. Instructions for System End Users
12. Conclusion
  1. Achievement
  2. Lessons Learned
  3. Acknowledgment
    - [Who financially supported your college study]
    - [Who sponsored your capstone project] Professor Jing Hou
    - [Who helped/encouraged you to achieve college success]
13. References
  - Appendix R
  - Appendix U
  - Appendix Agile Report

## Abstract

The rapid adoption of generative artificial intelligence has outpaced many users ability to interact with these systems effectively, creating a growing gap in prompt literacy. At California State University San Marcos, students and faculty are provided free access to the latest OpenAI models through an institutional license. However, many users are unsure how to write prompts that fully take advantage of these tools. This project proposes the development of a campus-wide prompt sharing website where students and faculty can create, explore, and share effective AI

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